



Rapid curing and self-stratifying lacquer coating with antifouling and anticorrosive properties

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ABSTRACT

Natural lacquers have long been used as environment-friendly and renewable coatings. Generally, they dry slowly and do not have fouling resistance limiting their applications. Herein, we report a rapid curing, self-stratifying amphiphilic hybrid lacquer coating with excellent antifouling and anticorrosive properties. The coating was prepared by using a natural lacquer coupled with hyperbranched polysiloxane terminated by multi-amine (HPSi) and silane-terminated amphiphilic telomer (S-FP) through a facile sol-gel process. The introduction of HPSi greatly shortened the drying time and improved the mechanical properties of the natural lacquer coating. As HPSi content increased, the drying speed and surface hardness of the coating increased. Particularly, the incorporation of a non-leaching, self-enriched amphiphilic telomer into the hybrid lacquer conferred it an excellent antifouling performance but preserved its anticorrosive and mechanical properties. The flexibility and bacteria-resistant ability of the coating were significantly improved with S-FP content. The study provides a new strategy for modification of natural lacquer, and the hybrid lacquers are expected to be used in development of coatings with both antifouling and anticorrosive performance.

1. Introduction

Natural lacquer is a type of natural polymer material with abundant resources that can be cured into a film under mild conditions through the catalytic oxidation of laccase [1]. Since the Neolithic age, natural lacquer has been used as a coating for handicrafts and industrial equipment because of its renewable and environmentally friendly nature. Furthermore, polymeric natural lacquer coating is very stable, durable, and impermeable, and exhibits excellent chemical resistance to a corrosive medium [2–8].

Generally, as a natural resin derived from lacquer sap, natural lacquer is composed of urushiol, glycoproteins, laccase enzymes, polysaccharides, and water [9]. Urushiol, the main film-forming constituent, is a catechol derivative with a straight-chain alkenyl group having a C15 chain and a typical urushiol is shown in Scheme 1. Laccase enzyme, another major ingredient, is the most important factor for the polymerization of the lipid components in lacquer sap. The polymerization mechanism of lacquer sap catalyzed by laccase was first reported in 1995; it is supposed to be accomplished in two stages including laccase-catalyzed polymerization and aerobic oxidative polymerization [10,11].

However, this procedure requires severe and specific conditions (70–90% relative humidity at 20–30 °C) for the activation of laccase, and a long drying time. Since then, some attempts have been made to accelerate the drying process of natural lacquer. For example, it is reported that the repeated-*kurome* process can be employed to accelerate the drying of natural lacquer, but this process takes a long time, and it is difficult to prepare the fast-drying natural lacquer in large quantities [12]. In addition, the prepolymerization of lacquer sap in a pure oxygen atmosphere can accelerate the drying of the lacquer film, but it is difficult to achieve [13]. Moreover, the ultraviolet irradiation method was employed to promote the drying of lacquer sap, but it severely damaged some properties of the lacquer coating, such as gloss and durability [14].

Chemical modification of natural lacquer is a better method to cure it quickly into a film under mild conditions. Urushiol, the key component of natural lacquer, is a catechol derivative with a linear alkenyl group, that has multiple active sites for chemical reactions. Generally, there are two methods to modify natural lacquer to accelerate its drying process: (1) the urushiol-metal coordination method and (2) the sol-gel method. Lin et al. prepared urushiol-metal polymers that could be dried quickly

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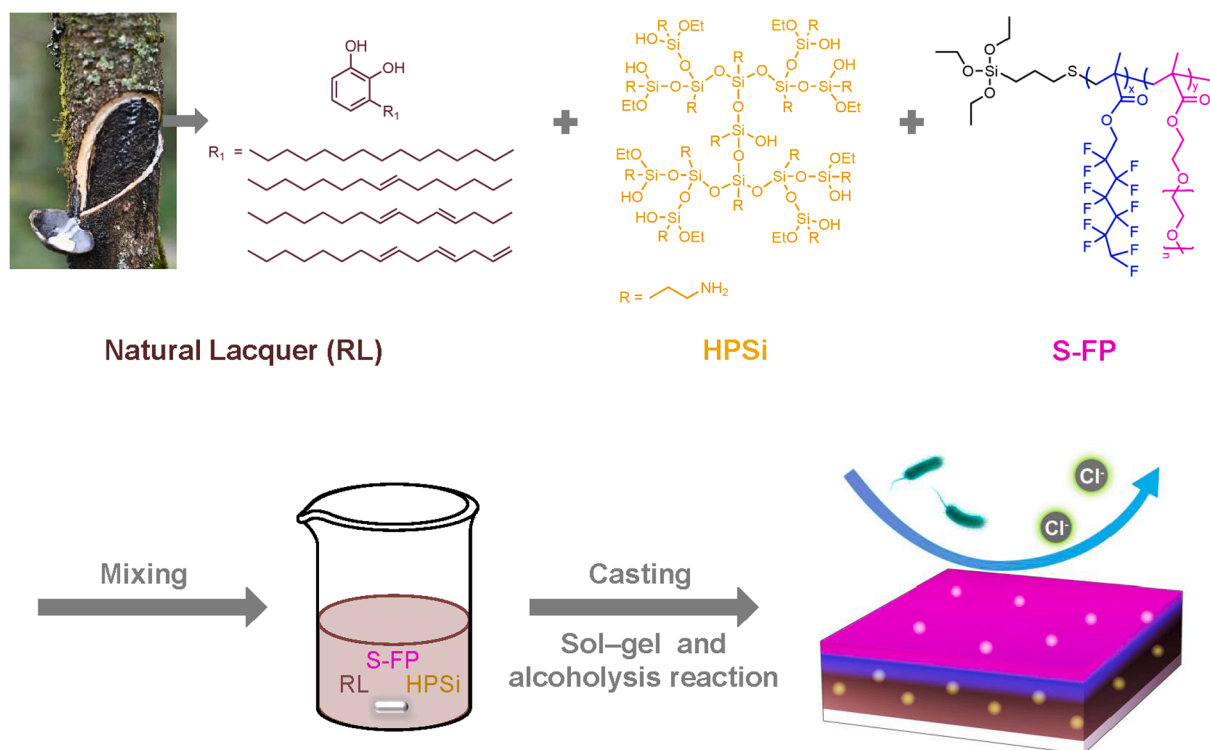
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Scheme 1. Schematic representation of the preparation of the hybrid lacquer coating.

at room temperature while exhibiting improved anticorrosion properties. However, the stability of the urushiol-metal coordination bond depended largely on the pH, and the metal ions had a negative impact on the laccase activity [15–17]. Alcoholysis between the urushiol hydroxyl group and the alkoxy group of the organic silanes, as well as the sol-gel reaction of the silanes, can promote the polymerization of urushiol; thus, natural lacquer modified with organic silanes has been reported to have quickly dried with good physical properties in a low-relative-humidity environment. The addition of organic silanes can also improve the anticorrosion and antiaging properties of the lacquer coating [18–21].

In addition to increasing the drying speed of natural lacquer, improving its antifouling abilities is a key research direction. As reported previously, natural lacquer exhibits excellent anticorrosion properties, but its antifouling abilities are limited, which limits its application in marine antifouling and other harsh environments [7,22–24]. Therefore, some researchers have added antibacterial nanoparticles or toxic metal substances to natural lacquer to improve its antifouling performance. It is reported that adding titanium dioxide nanoparticles to natural lacquer can increase the inhibition rate of the coating against *Escherichia coli* and *Staphylococcus aureus* [25]. However, titanium dioxide nanoparticles have weak resistance to bacteria, and they are prone to agglomeration. Furthermore, urushiol can be used as a reducing agent to prepare nano-silver in situ in polymer resins. When the content of nano-silver is only 0.3%, it has good antifouling performance [26,27]. However, the composite coating would release silver nanoparticles, which would cause short-term antifouling efficiency and pollute the marine environment.

In the present study, we have reported a rapid-curing, self-stratifying amphiphilic hybrid lacquer coating with excellent antifouling and anticorrosive properties. The coating was prepared using natural lacquer coupled with hyperbranched polysiloxane terminated by multi-amine (HPSi) and silane-terminated amphiphilic telomer (S-FP) via a facile sol-gel process. HPSi was synthesized and added to the natural lacquer to improve the drying process and mechanical properties, since HPSi has a highly branched structure, which has more highly reactive terminal amines and lower viscosity. S-FP was synthesized and added to the

natural lacquer to endow the coating with antifouling abilities. During the formation of the coating, the S-FP can be self-enriched on the surface, since the amphiphilic telomer S-FP with a low-surface-energy fluorocarbon segment is incompatible with the natural lacquer. We have systemically examined the properties of the coating in terms of its drying time, hardness, flexibility, adhesion, and surface wettability. Additionally, its antifouling and anticorrosive properties were also evaluated. Our aim is to develop a high-performance environmentally friendly bio-based coating with excellent antifouling and anticorrosive properties.

2. Materials and methods

2.1. Materials

Natural lacquer (RL) was purchased from Xi'an Raw Lacquer Coating Research Institute. DFMA from Harbin Xuejia Fluorin Silicon Chemical Co., Ltd. was used as received. PEGMA ($M_n = 475$ g/mol), KH550, and KH580 from Aladdin Industrial Corporation were used as received. Azobisisobutyronitrile (AIBN) from Aladdin was purified by recrystallization from methanol. Tetrahydrofuran (THF), anhydrous ethanol, and n-hexane were from Sinopharm and the THF was distilled by CaH_2 . The artificial seawater was prepared according to ASTM D1141-98 (2013).

2.2. Synthesis of HPSi

The synthetic procedure of HPSi has been reported elsewhere [28,29]. Under the protection of nitrogen, a mixture of KH550 (22.14 g, 100.0 mmol), water (2.16 g, 120.0 mmol), and ethanol (20.0 mL) were mixed and reacted at 60 °C for 4 h. After cooling to 25 °C, the mixture was concentrated under reduced pressure to remove the volatiles, and finally, HPSi was obtained as a colorless oil.

2.3. Synthesis of S-FP

The telomerization of KH580, PEGMA, and DFMA was conducted

Table 1

Drying time and mechanical properties of the hybrid lacquer coatings.

Samples	RL (g)	HPSi (g)	HPSi content (wt%)	Drying time (h, 25 °C, 60% RH)		Pencil hardness	Gloss	Impact resistance (cm)
				TF	HD			
RLH0	10	0	0	48	72	3H	58	3
RLH5	9.5	0.5	5	24	48	3H	75	20
RLH10	9	1	10	5	16	4H	54	15
RLH15	8.5	1.5	15	3	12	6H	19	8

TF: touch-free dryness, HD: hard dryness.

Table 2

Drying time and mechanical properties of the hybrid lacquer coatings.

Samples	RLH10 (g)	S-FP (g)	S-FP content (wt%)	Drying time (h, 25 °C, 60% RH)		Pencil hardness	Gloss	Impact resistance (cm)
				TF	HD			
RLH10	1	0	0	5	16	4H	54	15
RLH10S5	0.95	0.05	5	5	16	3H	46	25
RLH10S10	0.90	0.10	10	6	18	H	42	40
RLH10S15	0.85	0.15	15	6	20	2B	36	50

TF: touch-free dryness, HD: hard dryness.

following the procedure reported elsewhere [30,31]. KH580 (2.38 g, 10.0 mmol), PEGMA (9.50 g, 20.0 mmol), DFMA (8.00 g, 20.0 mmol), and AIBN (0.04 g, 0.2 wt%) were dissolved in THF (15.0 mL) and introduced into a Schlenk tube (100 mL) with a magnetic stir bar. The solution was deoxygenated by bubbling with nitrogen for 30 min and then reacted at 70 °C for 24 h. The resulting solution was purified by precipitation in hexane and dried in vacuum at 40 °C overnight. The telomers obtained were designated as S-FP.

2.4. Preparation of hybrid lacquer coatings

The hybrid lacquer coatings were prepared as follows. Natural lacquer, HPSi, and S-FP were dissolved separately in THF and then mixed and stirred for 20 min. The solution was cast onto an epoxy resin fiberglass panel (for surface wettability), steel (for the adhesion test), tinplate (for drying time, hardness, flexibility, impact resistance, and corrosion resistance tests), a glass slide (for the X-ray photoelectron spectroscopy test), and a silicon slide (for a bacteria-resistant assay). For a coating, the casting method is the most usual and efficient, which facilitates large-scale applications. The thickness was controlled by a tetrahedral wet film fabricator and determined by a coating thickness gage (Phynix, Model Surfux). Subsequently, all the coatings were cured at room temperature for at least 7 days to ensure complete curing. For convenience, the hybrid lacquer coatings were designated as RLHxSy according to the content of HPSi and S-FP. The specific composition can be seen in Table 1 and Table 2. The thickness of the coatings for images of the wrinkle phenomenon is ~ 400 µm.

2.5. Characterization

Proton Nuclear Magnetic Resonance Spectroscopy (¹H NMR). The chemical structure of HPSi and S-FP was characterized by ¹H NMR and ²⁹Si NMR spectra on a Bruker spectrometer (AV600 NMR). CDCl₃ was used as the solvent, and tetramethylsilane as the internal standard at room temperature.

2.6. Drying time

The drying properties of the hybrid lacquer coatings were evaluated according to the Chinese standard GB/T1728-1979.

2.7. Surface hardness test

The surface hardness of the coatings was subjected to a pencil

hardness test (QHQ-A, China) according to the standard of ISO 15184-2012 (Paints and varnishes–Determination of coating hardness by pencil test). The lacquer coatings were placed for 7 d to make it fully cured before the hardness was measured.

2.8. Impact resistance test

The impact resistance of the hybrid lacquer coatings was measured according to GB/T 1732-1993, where the higher the drop weight, the better the fracture toughness.

2.9. Flexibility test

The flexibility was tested according to the standard of ISO 1519-2011 (Paints and varnishes–Bend test–cylindrical mandrel).

2.10. Gloss test

The gloss was measured using a WGG60-Y4 gloss meter (Koshijia, China). Triplicate samples were used to calculate the average gloss.

2.11. Durability test

The durability of each coating was determined by a QUV accelerated weathering tester (Q-Panel Lab, Model QUV/spray). The coatings were deposited on the tinplate and fixed in the steel plate of the instrument. Then, the coatings were exposed to UV irradiation for 7 d, where the intensity was set at 0.83 W/m² and the temperature was set at 50 °C. After exposure, the gloss of coatings was determined again.

2.12. Abrasion-resistance test

The abrasion-resistance Tester (Guangzhou DMY Instrument, Model 339) was used to examine each coating. The coating was fixed on the board of tester by its clamps and covered with a piece of steel wool (1.5 cm × 1.5 cm). The position of the test rod with a flat bottom (1 cm × 1 cm) was adjusted until the bottom was in contact with the surface covered with the steel wool. Then, the rod with steel wool moved forward and backward on the surface of coating under a load of 200 g, where one forward and backward movement was regarded as a circle. The speed was set at 30 circles/min.

2.13. Moisture resistance test

The moisture resistance was tested by monitoring the mass change of the coating after 48 h in an environment of 35 °C and 90% relative humidity.

2.14. Adhesion test

The adhesion strength of the hybrid lacquer coatings was measured by a pull-off adhesion tester (PosiTest AT-A Automatic) according to ASTM D4541. The pull rate was set at 0.2 MPa/s and the value of each sample was calculated by averaging four data points.

2.15. Surface wettability measurements

The static water contact angle (WCA) of the hybrid lacquer coatings was measured on a Theta Auto 113 (KSV NIMA, Biolin Scientific) at room temperature by dropping 3 µL of deionized water using a sessile method. According to the Owens–Wendt–Rabel–Kaeble method, the surface energy (SE) of each sample could be calculated from the water and diiodomethane contact angles. The average contact angle value of each sample was calculated by averaging four data points.

2.16. X-ray photoelectron spectroscopy (XPS)

XPS analysis was used to investigate the surface compositions using an Axis Ultra DLD (Kratos Analytical) with a monochromate Al K α ($h\nu = 1486.6$ eV) source at takeoff angles of 90°.

2.17. Bacteria-resistant assays

The bacteria-resistant performance of the hybrid lacquer coatings was evaluated using the marine bacterial species *Pseudomonas* sp. nov 776, Gram-positive *S. aureus*, and Gram-negative *E. coli*, according to the previous procedure [26,32]. Typically, *S. aureus* was streaked on Trypticase Soy Agar (TSA) plates and grown at 37 °C for 24 h. Then, a monoclonal colony was inoculated into a tube containing 5.0 mL of Trypticase Soy Broth (TSB) solution and grown at 37 °C. Following the similar protocol, the marine bacteria *Pseudomonas* sp. and *E. coli* were cultured [33]. The suspension was diluted to 10⁸ cells/mL, and the hybrid lacquer coatings were immersed in a bacterial suspension (1 mL) and cultured for 4 h. Afterward, a LIVE/DEAD BacLight Bacterial Viability Kit was used to stain the bacteria that settled on the coating surface. The bacteria adhered onto the surface of the hybrid lacquer coatings were observed under a fluorescent microscope (Zeiss, Model Scope A1). The number of bacteria was determined by three random fields of view on triplicate slides. The long-term bacteria-resistant assays of the hybrid lacquer coatings were conducted according to the similar procedure.

2.18. Anti-biofilm assays

The anti-biofilm property of coatings was evaluated using the marine bacterial species *Pseudomonas* sp. nov 776, Gram-positive *S. aureus*, and Gram-negative *E. coli*. Typically, the samples were cast in 24-well plates and cured for 7 d to remove the residual solvent. Then, each coating was soaked in 1 mL of bacteria suspension (10⁸ cell/mL) and the plates were sealed by sealing films. The bacteria were incubated in an incubator at 30 °C. After 48 h, the suspension was removed and the plates were gently rinsed with ASW to take planktonic biofilm away. The biofilm was fixed by methanol and dried at room temperature. Afterwards, the fixed biofilm was stained with crystal violet (0.3 wt% aqueous solution) and rinsed with ASW to remove the extra dye. Finally, the dye was extracted with acetic acid (33 wt% aqueous solution) and measured by a photometer at 589 nm to obtain the OD value (optical density) as biomass.

2.19. Corrosion resistance measurements

The corrosion resistance of the hybrid lacquer coatings was investigated according to the previous literature [34–36]. To prevent corrosion occurring on the other surface of the tinplate sheet, a sealant (paraffin wax) was applied to the edges and back of the sample before testing. All the samples were immersed in 3.5 wt% of NaCl aqueous solution at 25 °C. The barrier performance was tested on a Zahner/Zennium potentiostat–galvanostat (Zahner, Germany) equipped with a common three-electrode cell arrangement. Generally, the coated tinplate sheet panel, Pt mesh, and saturated calomel electrode were used as the working electrode, counter electrode, and reference electrode, respectively. The frequency range of the electrochemical impedance spectroscopy (EIS) measurements was from 100 kHz to 10 mHz, and the amplitude perturbation was 100 mV. The polarization curves were obtained in the range of –250 to +250 mV versus the open-circuit potential (OCP) with a scan rate of 5 mV/s.

3. Results and discussion

3.1. Curing performance and mechanical properties of the hybrid lacquer coatings

Scheme 1 shows the chemical structures of urushiol, HPSi, and S-FP, and the preparation of the hybrid lacquer coating by enzymatic and sol–gel reactions. The ¹H NMR and ²⁹Si NMR spectra of HPSi are shown in Fig. S1 and S2. The data reveal that 3-aminopropyltriethoxysilane has been hydrolyzed and condensed into siloxane linkages. Moreover, the three different chemical environments of silicon atoms in HPSi correspond to terminal units (T), linear units (L), and dendritic units (D), respectively. The degree of branching of HPSi is calculated to be 0.95, indicating that HPSi has a highly branched structure. Actually, HPSi is close to perfect dendrimers with very small size [28–29,33]. Therefore, HPSi has not only better solubility and lower viscosity but also more highly reactive terminal amines than the linear analogs.

The drying time of the hybrid lacquer coatings containing different proportions of HPSi was examined, and the results are detailed in Table 1. The drying time of the natural lacquer coating (RLH0) was long at 60% RH and 25 °C, whose time of touch-free dryness (TF) and hard dryness (HD) was 48 and 72 h, respectively. With the increase of HPSi, the drying time was shortened remarkably, indicating that the curing speed of the hybrid lacquer coating was accelerated greatly. When the content of HPSi was 10.0 wt%, the hybrid lacquer coating dried very quickly and its time of TF and HD dryness was 5 and 16 h, respectively. The results clearly indicated that the polymerization rate of the hybrid lacquer was faster than that of the natural lacquer. As is known, natural lacquer is polymerized by enzymatic oxidation and auto-oxidation [13,37]. First, urushiol was catalyzed by laccase to form quinone radicals, and then radical transfer occurred to form dimers, trimers, and polymers. Furthermore, the carbon–carbon double bond on the side chain of the urushiol can undergo auto-oxidative polymerization, and can also react with urushiol quinone radicals to form a cross-linked structure. The shortening of the curing time of the hybrid lacquer coatings is due not only to the enzymatic reaction and auto-oxidation polymerization reaction, but also to the sol–gel reaction between the silane and the alcoholysis reaction between the natural lacquer hydroxyl groups and the silane. Moreover, HPSi has highly reactive terminal amines, so that it can undergo a Schiff base reaction and a Michael addition reaction with the urushiol quinone radicals, thereby further accelerating the polymerization [38–39]. As reported before [18], it is necessary to add a large amount of silane (~20 wt%) to the raw lacquer to achieve hard-drying in 36 h. However, due to the high reactivity of the hyperbranched silane, using less HPSi (~10 wt%) can achieve hard-drying in 16 h. On the other hand, the resulting fast-drying hybrid lacquer coating also has excellent physical and mechanical properties (Table 1). As the HPSi content increases, the pencil hardness of the

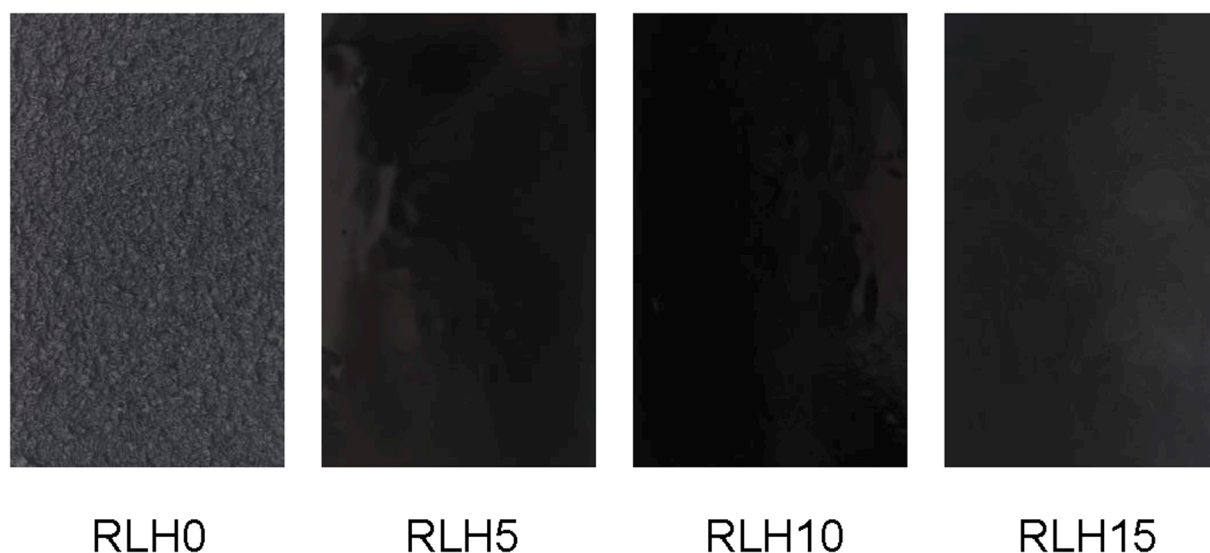


Fig. 1. Images of the wrinkle phenomenon of the hybrid lacquer coatings.



Fig. 2. Images of the 5-mm flexibility of the hybrid lacquer coatings.

hybrid lacquer coating increases gradually, but the flexibility exhibits a trend of first increasing and then decreasing. This is because the excessive addition of HPSi could make the cross-linking density of the natural lacquer coating too high, resulting in a decrease in flexibility. Nevertheless, the impact resistance of the natural lacquer coating with HPSi is better than that without HPSi. The gloss properties of the hybrid lacquer coatings exhibit the similar trend to the flexibility properties. Interestingly, the resulting hybrid lacquer coating can overcome the problem of wrinkling in thick coating. It is known that the polymerization mechanism of the natural lacquer is supposed to be accomplished in two stages including laccase-catalyzed polymerization and aerobic oxidative polymerization [10,11]. As the curing process of natural lacquer in the natural environment is very slow and requires the presence of oxygen, the curing speed of the surface of the natural lacquer coating will be faster than the curing speed inside, leading to the phenomenon of surface wrinkling when the coating thickness is large. The addition of HPSi increased the sol-gel and alcoholysis reaction between natural lacquer and silane, and the terminal amine of HPSi can undergo Schiff base reaction and Michael addition reaction with the urushiol quinone group, thereby accelerating the polymerization. Moreover, the addition of HPSi could increase the oxygen permeability of the coating, so when the surface layer is cured, the internal reaction will not slow down. Thus, the reaction inside is close to that of surface layer, making the coating have a smooth surface (Fig. 1).

Considering that the fast-curing hybrid lacquer coating has limited

antifouling abilities and cannot be used in various harsh environments, we added fouling resistant groups to improve its antifouling abilities. As reported before [30,31], PEG is an effective fouling-resistant polymer,

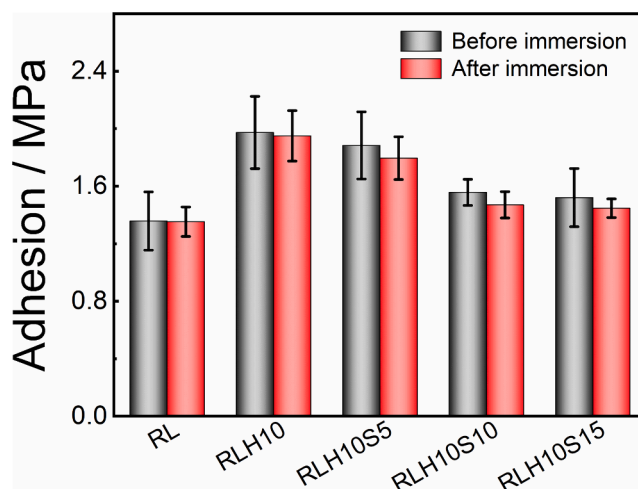


Fig. 3. Adhesion strength of the hybrid lacquer coatings on a steel substrate before and after immersion in water.

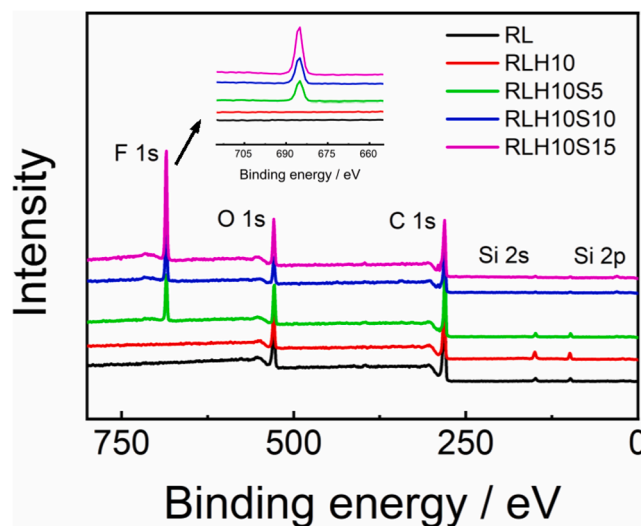


Fig. 4. XPS spectra of the hybrid lacquer coatings.

Table 3

The elemental mass content of F on the hybrid lacquer coatings.

Samples	F content in bulk	F content on the surface
RL	0 / 100	0 / 100
RLH10	0 / 100	0 / 100
RLH10S5	0.56 / 100	17.82 / 100
RLH10S10	1.14 / 100	28.56 / 100
RLH10S15	1.71 / 100	31.60 / 100

but PEG segments are readily buried in the bulk of the hybrid lacquer coatings since PEG segments have higher surface energy than that of hybrid lacquer. Thus, we synthesized an amphiphilic telomer (S-FP) and added it to the hybrid lacquer, since fluorocarbon segments have very low surface energy and are incompatible with hybrid lacquer, which can drag the PEG segments onto the surface. The successful synthesis of S-FP is confirmed by the ^1H NMR spectrum in Fig. S3, and its molecular weight (M_n) is ~ 2000 g/mol. The end group of the amphiphilic telomer is a silane coupling agent, which can undergo alcoholysis reaction with the phenolic hydroxyl group of the natural lacquer, and can also undergo a sol-gel reaction with HPSi. Taking RLH10 as a typical sample, we studied the effect of S-FP on the drying time and the mechanical, antifouling, and anticorrosion properties of the hybrid lacquer coating.

Table 2 shows the drying time and the mechanical properties of the hybrid lacquer coating containing different proportions of S-FP. The

images of the hybrid lacquer coatings in each period of the drying process were shown in Fig. S4. Clearly, the amphiphilic telomer S-FP has little effect on the drying time of the hybrid lacquer coatings. However, the addition of S-FP greatly reduced the surface pencil hardness and the gloss properties of the hybrid lacquer coatings. When 15% S-FP was added, the pencil hardness of the hybrid lacquer coating dropped from 4H to 2B. We also used the gloss to evaluate the durability. As shown in Fig. S5, although the gloss of RLH10 was lower than that of RL, it was reversed after the QUV accelerated weathering test for 7 d, which indicated that RLH10 possessed much better durability. On the other hand, the flexibility of the hybrid lacquer coating was improved significantly. When 15% S-FP was added, the impact resistance of the hybrid lacquer coating reached 50 cm without damage to the coating, indicating that the hybrid lacquer coating can also be used on the surface of flexible substrates for protection. The 5-mm flexibility test of the hybrid lacquer coating in Fig. 2 also supports this conclusion. Except for the RL coating, all the hybrid lacquer coatings can reach a flexibility of 5 mm. This is understandable that the addition of S-FP can increase flexibility, thus leading to significant improvement in impact resistance. In addition, the mass loss of all the hybrid lacquer coatings keeps within 4% after 400 cycles of the abrasion-resistance test (Fig. S6), indicating the hybrid lacquer coatings have excellent abrasion resistance. Besides, the hybrid lacquer coatings have excellent moisture barrier property (Fig. S7), and the moisture absorption rate of RLH10 is much lower than that of RL. This is because the addition of HPSi can increase the water

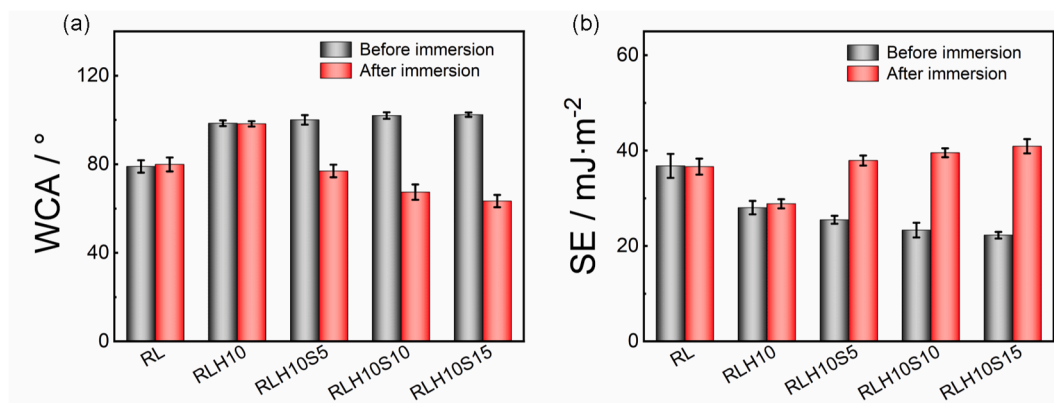


Fig. 5. (a) The WCA of the hybrid lacquer coatings before and after immersion in water for 24 h. (b) The SE of the hybrid lacquer coatings before and after immersion in water for 24 h.

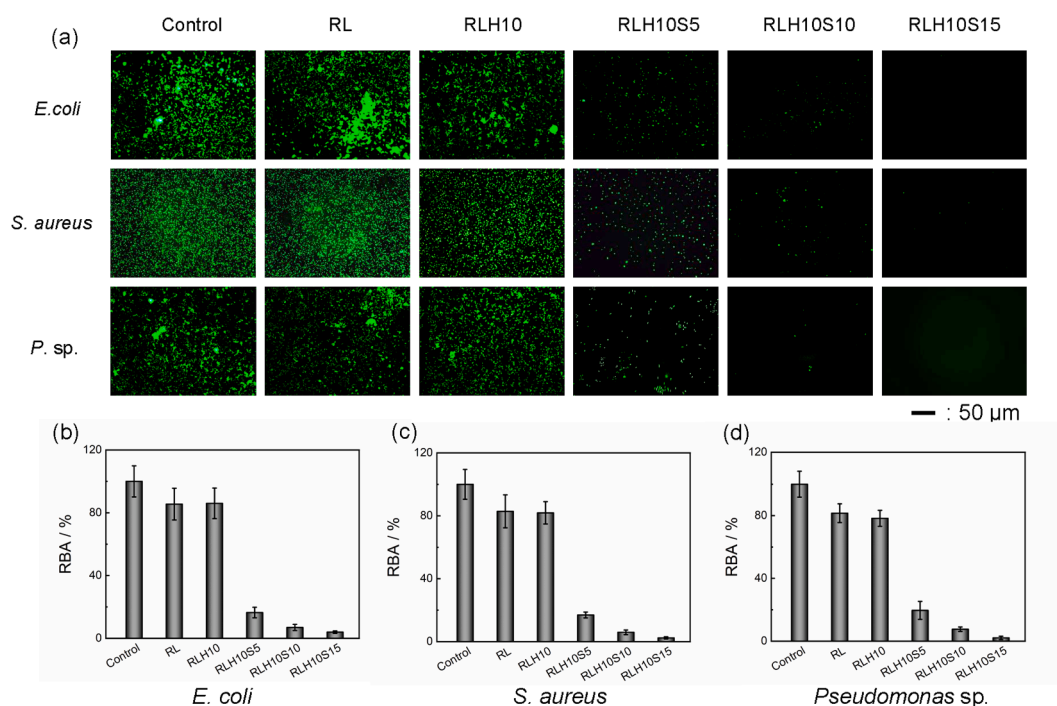


Fig. 6. (a) Fluorescence microscopy images of *E. coli*, *S. aureus*, and marine bacterial *Pseudomonas sp.* attached to the hybrid lacquer coatings when cultured for 4 h. Relative bacterial adhesion (RBA) of (b) *E. coli*, (c) *S. aureus*, and (d) marine bacterial *Pseudomonas sp.*

repellency of the coating.

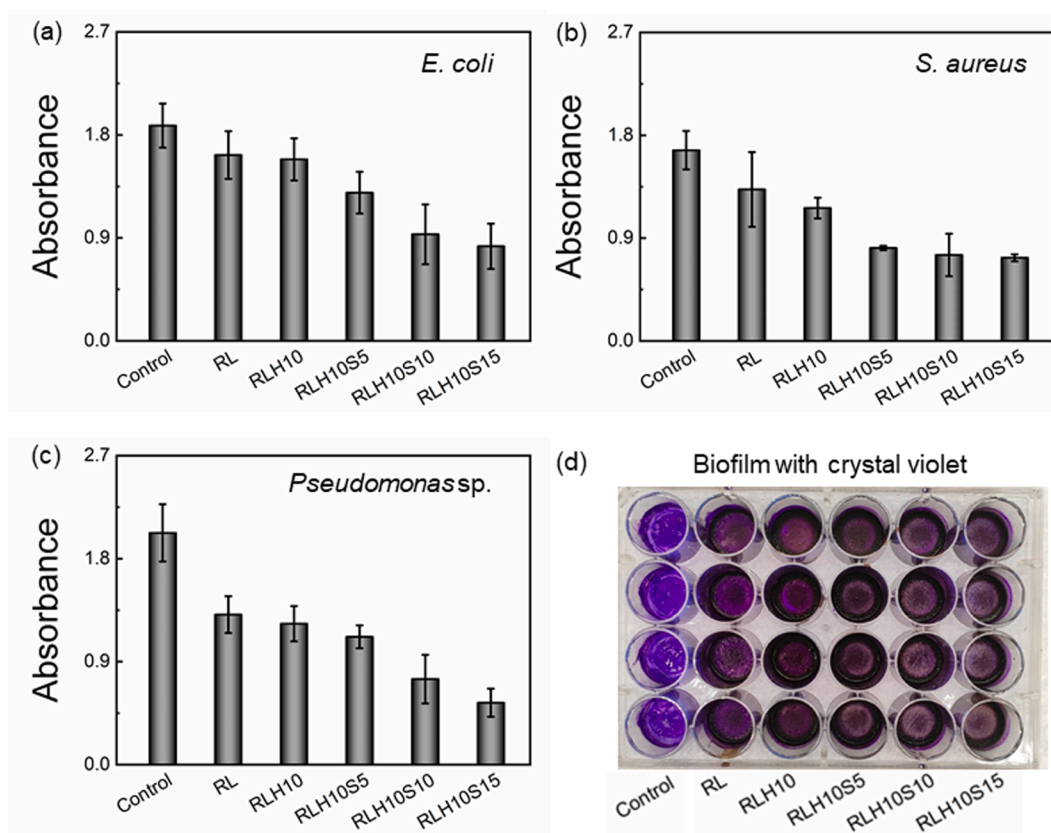


Fig. 7. The absorbance (optical density) of crystal violet residue in biofilm of (a) *E. coli*, (b) *S. aureus*, and (c) marine bacterial *Pseudomonas sp.* (d) Representative images of *Pseudomonas sp.* biofilm stained with crystal violet.

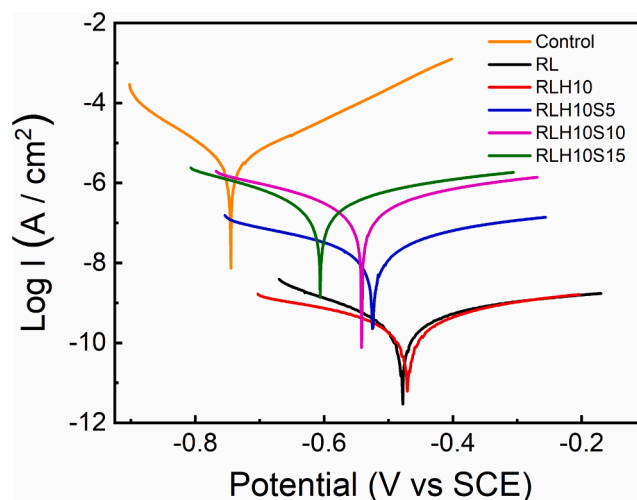


Fig. 8. Polarization curves of control (bare tinplate) and the hybrid lacquer coatings.

3.2. Adhesion strength of the hybrid lacquer coatings

Fig. 3 shows the adhesion strength of the hybrid lacquer coatings on a steel substrate before and after immersion in water. All the hybrid lacquer coatings exhibited an excellent adhesion strength higher than 1.3 MPa. This is because the natural lacquer contains abundant catechol groups that can form coordinate bonds with the metal substrate [40–42]. When HPSi was added to the natural lacquer, the adhesion increased, since there are hydrogen bonding and hydroxyl condensation between the silanol and the metal substrate. RLH10 has the highest adhesion strength of approximately 2.0 MPa. However, the adhesion of the fast-drying hybrid lacquer coating decreased slightly with the addition of the amphiphilic telomer S-FP. The adhesion strength of RLH10S5, RLH10S10, and RLH10S15 coatings are 1.88, 1.56, and 1.52 MPa, respectively. Increasing amphiphilic telomer content leads to a decrease in the hydrogen bonding interaction but an increase in the steric hindrance between hybrid lacquer polymer chains. Thus, both factors cause the coating to become weaker in adhesion and mechanical properties. Nevertheless, the adhesion strength was still higher than that of the RL coating. After immersion in water for 24 h, the adhesion of the RL and RLH10 did not change because of the hydrophobic nature. For the hybrid lacquer coatings containing S-FP, the adhesion decreased slightly, but was still higher than that of the natural lacquer.

3.3. Surface properties of the hybrid lacquer coatings

To verify the self-stratifying nature of amphiphilic oligomers, we observed the XPS spectra to examine the surface composition of the hybrid lacquer coatings. As shown in Fig. 4, the peak at 685.7 eV is attributed to F (1s) in the fluorocarbon groups, the peak at 533.1 eV to O (1s), the peak at 284.8 eV to C (1s), and the peaks at 150.5 and 100.4 eV are attributed to Si (2s and 2p). For the RL and RLH10 coatings, no F (1s) was detected because they do not contain any fluorocarbons. However, we observed the peak value of F(1s) for RLH10S5, RLH10S10 and RLH10S15. Moreover, the peak intensity of F (1s) increased with the increase of the S-FP content, as shown in Table 3. This is because of the incompatibility between fluorocarbon segments and hybrid lacquer as well as the very low surface energy of fluorocarbon. In other words, the amphiphilic telomer S-FP can be self-enriched on the surface during coating formation, so as to better exert the antifouling effect [43,44].

We also used the surface WCA and the SE of the hybrid lacquer coating before and after immersion to examine the surface self-enrichment of the amphiphilic telomer. As shown in Fig. 5, for the natural lacquer coating, the WCA was 79°. When 10% of HPSi was added, the WCA of the hybrid lacquer increased from 79° to 98°,

indicating that the hydrophobicity was increased significantly. As the S-FP content increased, the WCA of the hybrid lacquer coating increased slightly. However, after immersing in water for 24 h, the WCA of the hybrid lacquer coatings with S-FP decreased sharply. The reason for this is that the amphiphilic telomer S-FP contains a highly hydrophilic segment PEG, and when immersing in water, because of its strong hydrophilicity, it will in turn cover the fluorine surface. As a result, the surface is more hydrophilic, and the WCA of the RLH10S15 drops quickly from 102° to 63° after immersion in water for 24 h. Correspondingly, with the increase of the WCA of the hybrid lacquer coatings, the SE of the hybrid lacquer coatings decreases gradually because of the addition of low SE substances containing silicon and fluorine. After immersion in water for 24 h, as more PEG turns over to cover the fluorine surface, the SE of the hybrid lacquer coating also increases. That is, the change trend of the WCA and SE also confirms that the amphiphilic telomer S-FP will be self-enriched on the surface of the hybrid lacquer coating.

3.4. Fouling-resistant assays of the hybrid lacquer coatings

To evaluate the bacteria-resistant performance of the hybrid lacquer coatings, we performed bacteria-resistant assays against three kinds of bacteria: *E. coli*, *S. aureus*, and marine bacterial *Pseudomonas* sp. Fig. 6a shows the fluorescence microscopy images of the three kinds of bacteria attached to the modified lacquer coating, and Fig. 6b–d respectively show their RBA. A large number of live bacteria were observed on the control group sample made of silicon wafer, indicating that it is very susceptible to bacterial adhesion. The live bacteria were tightly adhered to the surface of the silicon wafers, and the distribution was very dense. For RL and RLH10, many bacteria were attached, proving that the natural lacquer has limited bacteria-resistant properties for *E. coli*, *S. aureus*, and marine bacterial *Pseudomonas* sp. The RBA rates of RL against *E. coli*, *S. aureus*, and marine bacterial *Pseudomonas* sp. were 85.5%, 81.5%, and 82.8%, respectively. However, as the S-FP content increased, the number of bacteria attached to the hybrid lacquer decreased, and the bacteria were almost invisible on the surface of RLH10S15. The bacteria adhered weakly to the surface of the hybrid lacquer coatings containing S-FP and were easily washed away by ASW or PBS. As we know, PEG is an effective fouling-resistant polymer that prevents the coating from adsorbing bacteria, instead of killing them [45–48]. The excellent bacteria-resistant performance is attributed to the surface enrichment of S-FP in the hybrid lacquer coating, where the hydrophilic PEG segment can form a hydration layer, thereby reducing the adhesion of bacteria. More importantly, the RBA rates of RLH10S15 against *E. coli*, *S. aureus*, and marine bacterial *Pseudomonas* sp. can reach 3.9%, 2.1%, and 2.4%,

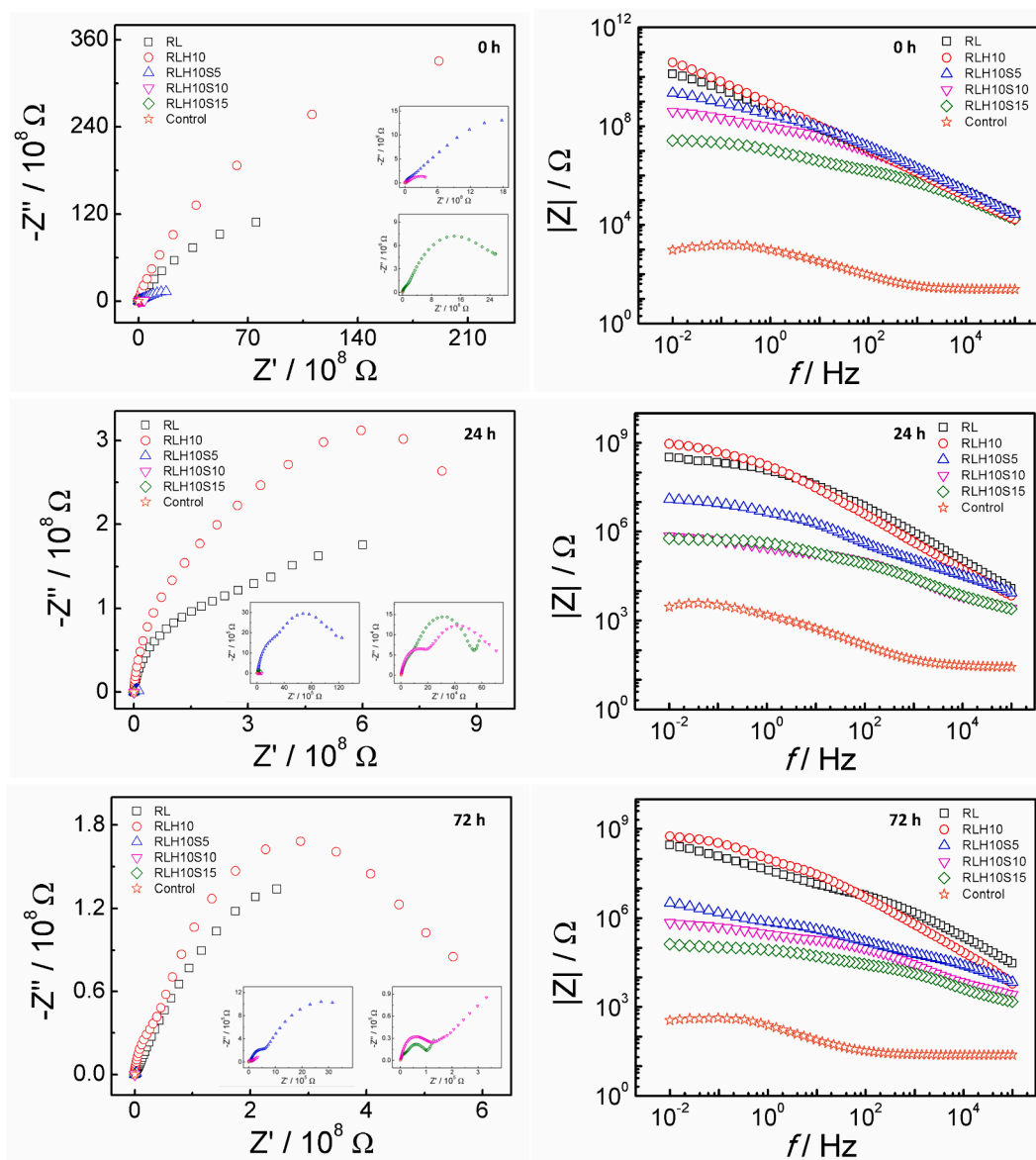


Fig. 9. EIS responses in Nyquist and Bode plots of the hybrid lacquer coatings after immersion in 3.5 wt% NaCl aqueous solution.

Table 4

Corrosion parameters determined from Tafel curves of control (bare tinplate) and the hybrid lacquer coatings.

Samples	OCP/V	E_{corr}/V	$I_{corr}/(A/cm^2)$	$R_p(\Omega)$
Control	-0.627	-0.742	3.108×10^{-6}	6.050×10^5
RL	-0.417	-0.478	3.318×10^{-10}	2.424×10^9
RLH10	-0.398	-0.471	2.601×10^{-10}	3.064×10^9
RLH10S5	-0.523	-0.525	1.836×10^{-8}	4.696×10^7
RLH10S10	-0.527	-0.539	2.034×10^{-7}	4.395×10^6
RLH10S15	-0.546	-0.608	2.967×10^{-7}	2.620×10^6

respectively, indicating that the bacteria-resistant properties of hybrid lacquer coatings have a broad spectrum. The long-term bacteria-resistant test is also conducted, and the hybrid lacquer coatings still have excellent antibacterial properties (Fig. S8 and Fig. S9). Therefore, the hybrid lacquer coating can be used widely as a protective coating in various environments and fields.

Fig. 7 shows the attachment of three kinds of bacteria: *E. coli*, *S. aureus*, and marine bacterial *Pseudomonas* sp. biofilm on the hybrid lacquer coatings. The absorbance of the RL and RLH10 are slightly lower

than that of the control, which indicates that they have poor resistance to biofilm. As the S-FP content increases, the absorbance of the hybrid lacquer coatings decreases, indicating the anti-biofilm ability improves. The inhibition of biofilm growth further indicates that the telomer S-FP is enriched on the surface, which makes the hybrid lacquer coating exhibit excellent antifouling ability.

3.5. Anticorrosive performance of the hybrid lacquer coatings

The anticorrosive performance of the hybrid lacquer coatings was investigated by means of a series of electrochemical measurements [7,49]. Open-circuit potential (OCP) measurements were first performed to ensure steady-state conditions. Then, the EIS measurements and the polarization curves were carried out in a 3.5 wt% NaCl aqueous solution and the results were shown in Fig. 8 and Fig. 9. Values of some important parameters such as corrosion potential (E_{corr}), corrosion current densities (I_{corr}) and polarization resistance (R_p) were calculated from the Tafel polarization curves and summarized in Table 4. Note that the polarization resistance (R_p) values were calculated from the Tafel plots, according to the Stern–Geary equation (eqn (1)) [50].

$$R_p = \frac{b_a b_c}{2.303(b_a + b_c)I_{corr}} \quad (1)$$

where I_{corr} is determined by the intersection of the linear part of the anode curve and the cathode curve, and b_a and b_c are the anodic and cathodic slopes, respectively.

Generally, a higher E_{corr} , a lower I_{corr} and a higher R_p indicate better corrosion protection performance. As shown in Fig. 8 and Table 4, the control (bare tinplate) had the lowest E_{corr} of -0.742 V and the highest I_{corr} of 3.108×10^{-6} A/cm², indicating the worst corrosion resistance. The RL had E_{corr} of -0.478 V and I_{corr} of 3.318×10^{-10} A/cm², and RLH10 had E_{corr} of -0.471 V and I_{corr} of 2.601×10^{-10} A/cm², which indicated that RL and RLH10 had excellent corrosion resistance. The R_p values of RL and RLH10 were $2.424 \times 10^9 \Omega$ and $3.064 \times 10^9 \Omega$, respectively, which further confirmed their good barrier performance. As the S-FP content increases, the value of I_{corr} increases, and the R_p value decreases, indicating that the corrosion resistance becomes worse. Anyhow, the R_p value of RLH10S15 is still much higher than that of bare tinplate.

Fig. 9 shows the corrosion resistance of the hybrid lacquer coatings evaluated using EIS measurements. Generally, the corrosion resistance can be analyzed by the capacity reactance arc radius and the impedance value [51]. In the initial stage of immersion, all the coatings have a high impedance value of over $10^7 \Omega$. As the S-FP content increases, the impedance value of the coating decreases, indicating that the corrosion resistance of the coating decreases. Nevertheless, the impedance value of RLH10S15 is still much higher than that of bare tinplate. From the Nyquist plots, only a semi-circular arc with high impedance can be clearly observed, which is probably because some water has penetrated into the coating, but has not reached the coating/metal interface. The experimental data can be fitted by the equivalent circuit model (Fig. S10a). In this circuit, R_s stands for the solution resistance, Q_c is the coating capacitance, and R_c is the coating resistance [52–53].

After immersion for 24 h, the second time constant can be observed in the Nyquist plots of RLH10S5 and RLH10S10, indicating that the electrolyte solution has partially penetrated the coating and the metal substrate has begun to corrode. The experimental data can be fitted by the equivalent circuit model (Fig. S10b), in which Q_{dl} and R_t are the electric double layer capacitance and the charge transfer resistance of the metal corrosion reaction, respectively. Note that the Nyquist plot of RLH10S15 has a Warburg impedance diffusion tail, since more and more corrosion products are deposited and covered on the metal surface, making the interfacial corrosion reaction be controlled by the mass transfer diffusion process. The experimental data can be fitted by the equivalent circuit model (Fig. S10c), in which Z_w represents the Warburg diffusion resistance.

After immersion for 72 h, the second time constant can be observed in the Nyquist plot of RLH10S5, and the Warburg impedance diffusion tail can be observed in the Nyquist plots of RLH10S10 and RLH10S15, which indicates that the corrosion resistance of the coating has deteriorated with the extension of the immersion time. However, RL and RLH10 still kept a semi-circular capacitive reactance arc, indicating excellent corrosion resistance of RL and RLH10. This is probably because RL and RLH10 have a higher crosslink density and a denser structure.

4. Conclusions

We have developed a novel rapid curing hybrid lacquer coating via a facile sol–gel reaction and an alcoholysis reaction. The coating can not only cure quickly in a mild condition but also achieve thick coating without wrinkling. The coating exhibits high adhesion and excellent mechanical properties, with good flexibility and high hardness. In particular, the coating exhibits excellent fouling-resistant performance toward various bacteria, because of the self-enrichment of the amphiphilic telomer. Meanwhile, the coating still maintains high barrier performance and corrosion resistance. This study provides a new strategy

for modification of natural lacquer, and the hybrid lacquers are expected to be used in development of coatings with both antifouling and anti-corrosive performance.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2021.129755>.

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